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Day 1

STARTING SOON

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Day 1

Welcome to Data Visualization Camp

Class begins shortly — please take a seat

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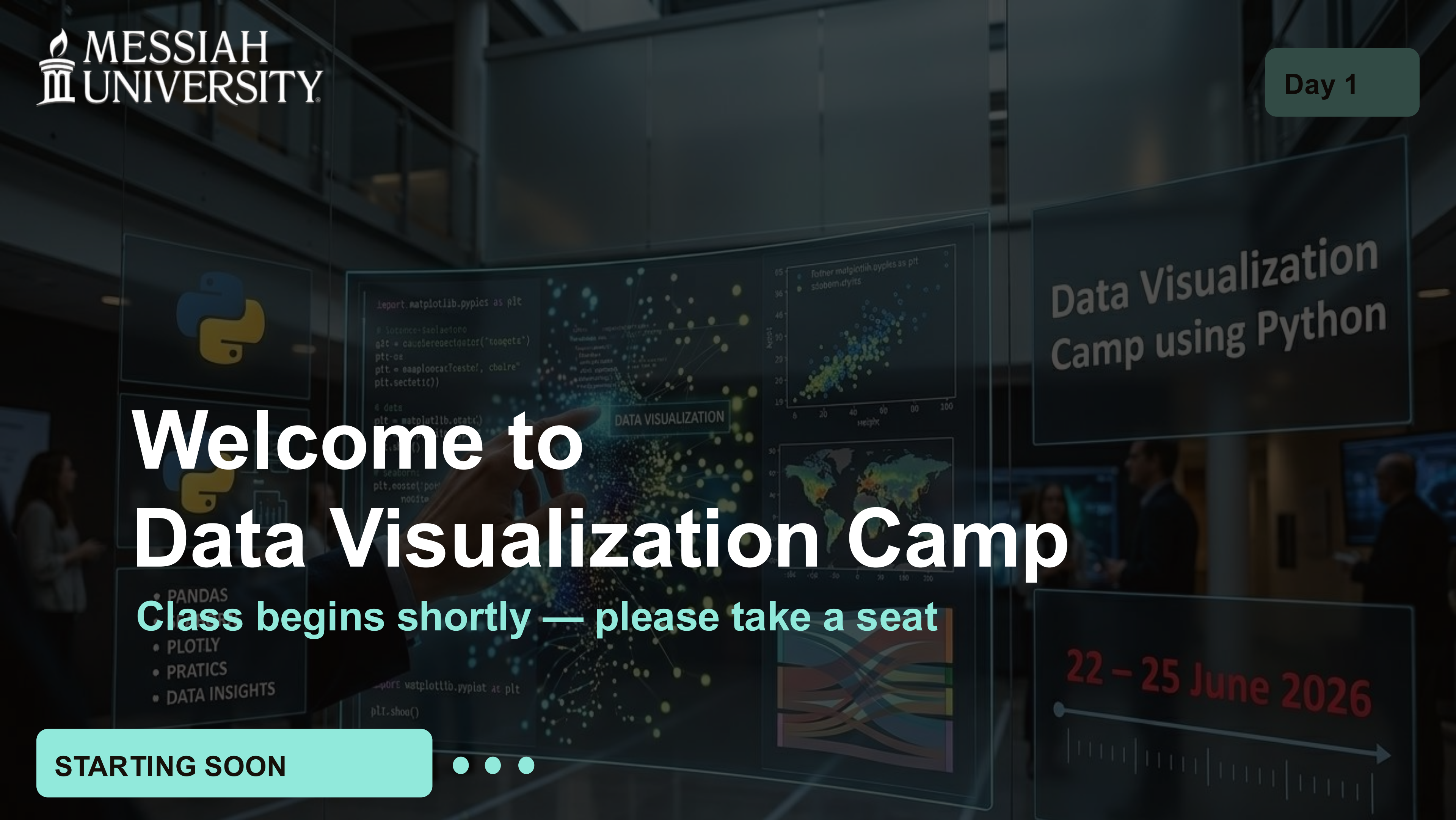

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Let’s Know Each Other!!!

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What You'll Learn

- Setup & ‘Hello World’
- Choosing Your Visual Story
- Role of Data in Science
- Matplotlib Basics
- Guided Lab

Data Visualization Camp using Python

22 – 25 June 2026

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🛠 Meet Your Tools

🐍 Python

A beginner-friendly programming language

Used by: NASA, Netflix, Instagram, Google

Powers: AI, Data Science, Automation

Why Python? Simple syntax — reads almost like English!

☁ Google Colab

A FREE notebook in your browser

No installation needed — just a Google account

Write + run Python in the same place

Perfect for data science & visualization!

colab.research.google.com

≡ Colab | Untitled.ipynb

Connect

File Edit View Insert Runtime Tools Help

+ Code + Text

```
Type in your browser:  colab.new

(or: colab.research.google.com)
```

Step 1 of 3: Open Google Colab

👆 **In your browser address bar, type: colab.new and press Enter**

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colab.research.google.com

≡ Colab | Untitled.ipynb

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```
# Click the first empty cell below
# You will see a blinking cursor
# That means it is ready — type your code there!
```

Step 2 of 3: Create a New Notebook

👆 **Click the cell with [ ] on the left — that green ▶ button runs your code!**

colab.research.google.com

≡ Colab | Untitled.ipynb

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+ Code + Text

```python
print("Hello, World!")
```

```
Hello, World!
```

Step 3 of 3: Execute

Type print("Hello, World!") then press the green ▶ button or Shift+Enter

Variables — Giving Names to Data

Think of a variable like a labeled box — you put a value in and give it a name!

```python
# Number (integer)
age = 16

# Decimal (float)
gpa = 3.85

# Text (string)
name = "Alex"

# True or False (bool)
loves_pizza = True
```

`int`
Whole numbers — e.g. 17, -3, 100

`float`
Decimal numbers — e.g. 3.14, 98.6

`str`
Text / words — e.g. "hello", "data"

`bool`
True or False — e.g. True, False

💡 Pro tip: Use snake_case for names with multiple words — e.g. my_score = 95

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Python as a Calculator

```python
# Basic math operations
5 + 3
# Output: 8

10 - 4
# Output: 6

6 * 7
# Output: 42

15 / 4
# Output: 3.75

2 ** 8
# Output: 256 (power!)
```

Operator	Operation	Example
+	Addition	3 + 4 = 7
/	Division	9 / 2 = 4.5
-	Subtraction	10 - 3 = 7
**	Power	2**10 = 1024
*	Multiply	5 * 6 = 30
%	Remainder	17 % 5 = 2

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input() & print() — Making Python Talk

print() — Show Output

```python
# Print text
print("I love data!")

# Print a variable
score = 99
print("Score:", score)

# Print multiple things
name = "Alex"
print("Hi,", name, "!")
```

input() — Get User Input

```python
# Ask for a name
name = input("Your name? ")

# Ask for a number
age = int(input("Your age? "))

# Use both together
print("Hello,", name)
print("You are", age, "years old")
```

Python Libraries — Superpowers!

Libraries are like apps for Python. One line of code and you get 1000s of new features!

matplotlib

Draw charts, graphs & plots

```
import matplotlib.pyplot as plt
```

seaborn

Beautiful statistical charts

```
import seaborn as sns
```

pandas

Work with tables & data files

```
import pandas as pd
```

numpy

Math & arrays at super speed

```
import numpy as np
```

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🏆 Mini Challenge: Your Profile Program

📋 Your Mission:

1. Ask for user's name
2. Ask for their age
3. Ask their fav color
4. Print a fun greeting

BONUS:

- Calc year they were born

⏱ **10 minutes — Try it yourself, then share with the class!**

🏆 Mini Challenge: Your Profile Program

📋 Your Mission:

1. Ask for user's name
2. Ask for their age
3. Ask their fav color
4. Print a fun greeting

BONUS:

- Calc year they were born

```
# My Profile Program

name = input("Your name? ")
age  = int(input("Age? "))
color= input("Fav color? ")

# Your print statements here...
print("Hello,", name, "!")

# BONUS: calculate birth year
birth_year = 2026 - age
print("Born in:", birth_year)
```

⏱ **10 minutes — Try it yourself, then share with the class!**

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Thanks!